Wafer Probe Yield Monitoring and Defect Pareto Analysis on the WM-811K Dataset

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Abstract

Wafer probe maps record which dies on each wafer pass or fail electrical test, and are a primary input to yield reviews in semiconductor assembly and test. This study analyzes 811,457 wafer maps across 46,293 lots from the public WM-811K dataset. Die-weighted yield is 93.3%. Because lot-to-lot variation is 31 times larger than binomial sampling noise, a standard p-chart flags 97% of lots and is not usable, while a Laney p′ chart flags 1,716 high-fail lots (3.7%) that hold 24.2% of all dies lost. Among labeled wafers, Edge-Ring is the largest patterned loss (47.9%). However, labeled wafers hold only 6.1% of all dies lost and differ in yield distribution from unlabeled wafers (Kolmogorov-Smirnov statistic 0.35). A gradient-boosted classifier, evaluated on held-out lots at 87.4% macro recall, labels the remaining 638,507 wafers. The resulting Pareto ranks Near-full (37.8%) and Random (17.3%) above Edge-Ring (15.2%). A model-free check confirms that near-dead wafers are 5.5 times more common among unlabeled wafers. In this dataset, lot excursion detection and containment is a larger lever than chronic edge loss.

1. Introduction

Yield reviews in wafer probe and packaging programs answer three practical questions. Which lots are abnormal and need investigation, what kinds of defects account for the most lost dies, and how far can the available labels be trusted. This study builds a reproducible pipeline that answers each question on a large public dataset, and tracks how the answer to the second question changes once the limits of the labels are taken into account.

The analysis proceeds in four stages, each building on the last.

- Measure wafer and lot yield with consistent, die-weighted definitions.
- Monitor lot fail rate with a control chart suited to large lots, and rank flagged lots by excess fails.
- Rank defect patterns by dies lost, test whether labeled wafers represent the full dataset, and relate patterns to flagged lots.
- Train a defect pattern classifier, label every unlabeled wafer, and recompute the Pareto over all yield loss.

2. Data

WM-811K [1] contains 811,457 wafer maps from real fabrication. Each map is a grid in which 0 marks a position with no die, 1 a passing die, and 2 a failing die. Maps come in 632 distinct grid shapes, reflecting different products. 172,950 wafers (21.3%) carry one of nine labels, eight defect patterns plus *none*. The dataset includes lot identifiers but no timestamps or product identifiers. Table 1 summarizes its scale.

Table 1. Dataset summary.

Quantity	Value
Wafers	811,457
Lots	46,293
Median wafers per lot	24
Tested dies	1,493,891,189
Labeled wafers	172,950 (21.3%)
Distinct map shapes	632

3. Methods

3.1 Yield definitions

Wafer yield is passing dies divided by tested dies on one wafer. Lot yield is total passing dies divided by total tested dies across the lot, so larger wafers carry more weight. Overall yield is computed the same way over the full dataset. Wafers with no tested dies are excluded from yield and counted separately.

3.2 Statistical process control

Each lot i has fail rate p_i over n_i dies, and the pooled fail rate is p_{avg}. A standard p-chart sets limits at

$$p_{\text{avg}} \pm 3\sigma_i, \quad \sigma_i = \sqrt{(p_{\text{avg}}(1 - p_{\text{avg}}) / n_i)}$$

which assumes all variation between lots is binomial sampling noise. With tens of thousands of dies per lot, those limits are very narrow. The Laney p′ chart [2] converts each lot to $z_i = (p_i - p_{\text{avg}}) / \sigma_i$, estimates their spread from the average moving range of consecutive z-scores, and widens the limits by that factor.

$$\sigma_z = \text{mean}|z_i - z_{i-1}| / 1.128, \quad \text{limits } p_{\text{avg}} \pm 3\sigma_i\sigma_z$$

A value of σ_z near 1 indicates purely binomial variation. Because the moving range depends on lot order and the data has no timestamps, lots are ordered by the number in their lot name. For each flagged lot, excess fails are the observed fails minus $p_{\text{avg}} n_i$, the fails expected at the average rate.

3.3 Pareto, representativeness, and lift

Patterns are ranked by total dies lost rather than by wafer count, since a rare pattern can cost more dies than a common one. To test whether labeled wafers represent the dataset, their wafer yield distribution is compared with that of unlabeled wafers using the two-sample Kolmogorov-Smirnov (KS) statistic, the largest gap between the two empirical distribution functions. To relate patterns to the control chart, each pattern's share among labeled wafers in flagged lots is divided by its share in all other lots. This ratio, called lift, exceeds 1 when a pattern is over-represented in flagged lots.

3.4 Defect pattern classifier

Each wafer map is converted to 85 engineered features so that maps of any shape enter one model. The features are the overall fail rate, fail rates in an 8 by 8 grid, in five concentric rings, and in eight angular sectors, the ratio of edge to overall fail rate, the mean and spread of failing dies' radial position, and

statistics on connected clusters of failing dies, including how elongated the largest cluster is. A histogram-based gradient-boosted tree classifier is trained with balanced class weights, because *none* makes up 85% of labeled wafers. Train and test sets are split by lot, 80% and 20%, so no lot contributes wafers to both. Results are reported as per-class precision and recall and as macro recall, the unweighted mean of per-class recall.

A final model trained on all labeled wafers then predicts a pattern and a confidence, the top class probability, for each unlabeled wafer. The all-wafer Pareto uses true labels where they exist and predictions elsewhere, and counts predictions below 0.8 confidence as *uncertain*.

4. Results

4.1 Yield overview

Overall die-weighted yield is 93.3%. Median wafer yield is 92.9% and median lot yield is 92.4%. Overall yield sits above both medians, which is consistent with the positive relationship between lot size and lot yield (Spearman rank correlation 0.34). Larger lots carry more weight in the die-weighted figure and also tend to yield better. Figure 1 shows the lot yield distribution.

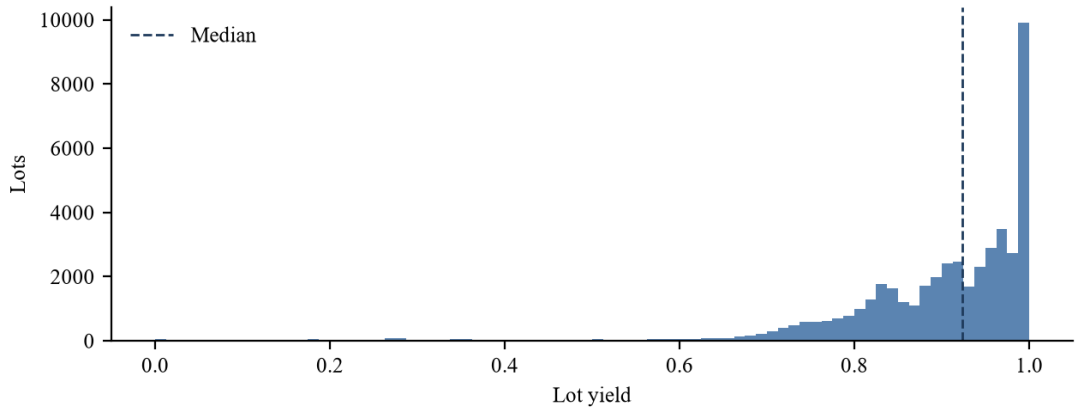


Figure 1. Distribution of lot yield across all lots. The dashed line marks the median.

4.2 Lot monitoring

The estimated σ_z is 31.3, so lot-to-lot variation is about 31 times what binomial noise alone would produce. At the median lot size of 18,156 dies, the standard chart's limits extend only ±0.56 percentage points from the center line, and it flags 96.6% of lots on one side or the other. The Laney chart flags 1,716 lots on the high side, about 27 times the 62 expected by chance in a stable process (Table 2). These lots hold 24.2% of all dies lost, and their excess fails total 21.1 million dies, or 21.0% of all yield loss.

Table 2. Standard p-chart and Laney p′ chart on lot fail rate.

	Standard p-chart	Laney p′ chart
Lots flagged high	23,558 (50.9%)	1,716 (3.7%)
Lots flagged low	21,151	666
Median limit half-width	0.56 pp	17.45 pp
Share of dies lost in high-flagged lots	72.4%	24.2%

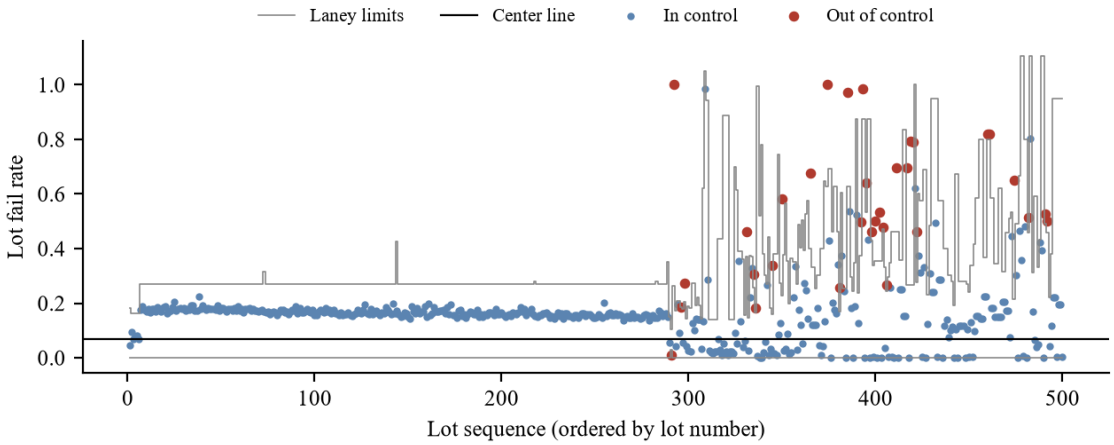


Figure 2. Laney p′ chart for the first 500 lots by lot number. Limits differ by lot because each lot has a different die count.

4.3 Defect Pareto on labeled wafers

Among labeled wafers with a pattern, Edge-Ring accounts for 47.9% of dies lost, and 4 patterns reach 80% (Table 3). Ranking by dies lost and ranking by count disagree for low-frequency, high-severity patterns. Random wafers lose 550 dies each and Scratch wafers 129.

Table 3. Patterned dies lost among labeled wafers, ranked by dies lost.

Pattern	Wafers	Dies lost	Dies lost per wafer	Share	Cumulative
Edge-Ring	9,680	2,934,194	303	47.9%	47.9%
Edge-Loc	5,189	970,907	187	15.8%	63.7%
Center	4,294	732,198	171	12.0%	75.7%
Loc	3,593	530,656	148	8.7%	84.4%
Random	866	475,955	550	7.8%	92.1%
Donut	555	220,197	397	3.6%	95.7%
Scratch	1,193	154,388	129	2.5%	98.2%
Near-full	149	107,249	720	1.8%	100.0%

4.4 Coverage and representativeness of labels

Labeled wafers with a pattern hold only 6.1% of all dies lost. Labeled *none* wafers hold 12.1% and unlabeled wafers 81.8%. The two groups also differ in yield (Table 4). The KS statistic is 0.35.

Unlabeled wafers are more spread out, with more near-perfect wafers and more very poor ones, so the labeled Pareto in Table 3 describes the labeled set and should not be read as a dataset-wide result.

Table 4. Wafer yield of labeled and unlabeled wafers.

Group	Wafers	10th pct.	Median	90th pct.	Share below 80%
Labeled	172,950	80.1%	89.4%	95.7%	9.7%
Unlabeled	638,507	78.2%	94.7%	99.9%	12.1%

4.5 Patterns in flagged lots

In flagged lots, 77% of labeled wafers carry a defect pattern, compared with 13% in other lots. The most over-represented patterns are Random (57.2 times), Donut (29.8 times), Near-full (8.4 times) (Table 5). Scratch is under-represented (0.33). Part of this effect is mechanical, since a few high-severity wafers can push a lot above its limit by themselves. Lift therefore reflects severity as well as association. Only 13.2% of wafers in flagged lots are labeled, and estimates for rare patterns rest on few wafers.

Table 5. Pattern mix among labeled wafers in Laney-flagged lots and in all other lots.

Pattern	Wafers in flagged lots	Share, flagged	Share, other	Lift
Random	491	12.7%	0.2%	57.20
Donut	225	5.8%	0.2%	29.79
Near-full	24	0.6%	0.1%	8.39
Center	645	16.7%	2.2%	7.72
Edge-Ring	1,152	29.8%	5.0%	5.90
Edge-Loc	323	8.3%	2.9%	2.90
Loc	112	2.9%	2.1%	1.41
Scratch	9	0.2%	0.7%	0.33
none	889	23.0%	86.7%	0.27

4.6 Classifier performance

On 34,947 wafers from held-out lots, macro recall is 87.4% and accuracy is 94.0% (Table 6, Figure 3). Edge-Ring is recognized with 96% recall at 98% precision. Precision is lowest for Scratch (33%), Loc (39%), Edge-Loc (60%). These patterns are localized, and because *none* wafers are so numerous, even a small share of them predicted as a localized pattern outnumbers the true cases. Balanced class weights make this trade, accepting false alarms in exchange for recall on rare patterns.

Table 6. Classifier results on held-out lots.

Pattern	Precision	Recall	F1	Test wafers
Center	0.81	0.95	0.87	904
Donut	0.87	0.90	0.88	167
Edge-Loc	0.60	0.84	0.70	1,013
Edge-Ring	0.98	0.96	0.97	1,947
Loc	0.39	0.76	0.52	674
Near-full	1.00	0.97	0.98	33
Random	0.89	0.83	0.86	147
Scratch	0.33	0.72	0.45	230
none	1.00	0.95	0.97	29,832

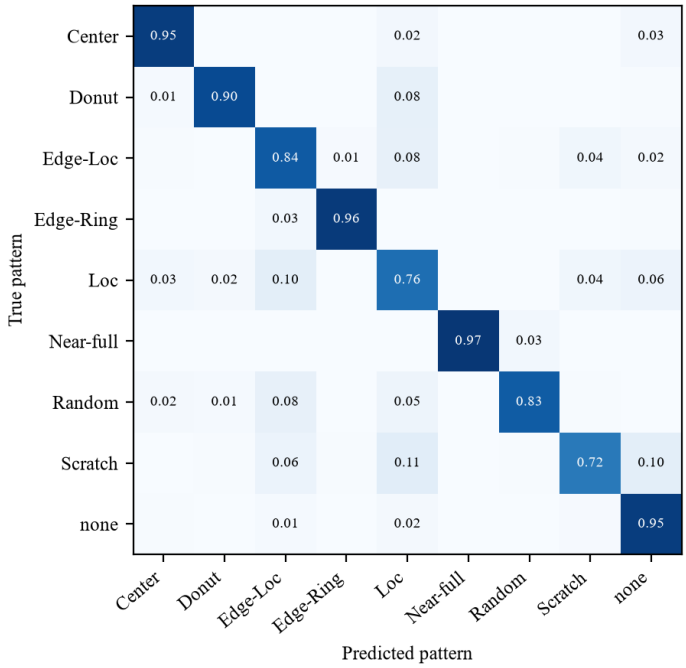


Figure 3. Confusion matrix on held-out lots, normalized by row so the diagonal shows recall for each true pattern.

4.7 Pareto over all wafers

Combining true labels with confident predictions changes the ranking (Table 7, Figure 4). Near-full rises to the largest patterned loss at 37.8%, followed by Random at 17.3%. Together they account for 55.0%. Edge-Ring falls to third at 15.2%. Predictions below 0.8 confidence account for 16.3% of all dies lost and are excluded from the shares. The top three patterns are also the ones the classifier predicts most precisely, so the reordering does not depend on the weak classes.

Because the Near-full result depends on the classifier, it was checked against a rule that does not use the model. Wafers below 20% yield number 2,310 among unlabeled wafers (0.36%) and 113 among labeled wafers (0.065%), a ratio of 5.5. Near-dead wafers were largely left unlabeled, which explains why the labeled Pareto underweights them.

Table 7. Patterned dies lost over all wafers, using true labels where available and predictions with confidence of at least 0.8 elsewhere.

Pattern	Wafers	Of which predicted	Dies lost	Share	Cumulative
Near-full	3,160	3,011	13,260,032	37.8%	37.8%
Random	8,003	7,137	6,070,235	17.3%	55.0%
Edge-Ring	15,676	5,996	5,329,000	15.2%	70.2%
Edge-Loc	21,996	16,807	4,045,041	11.5%	81.7%
Center	20,152	15,858	3,165,505	9.0%	90.8%
Loc	76,526	72,933	1,548,180	4.4%	95.2%
Scratch	10,247	9,054	1,357,247	3.9%	99.0%
Donut	773	218	340,211	1.0%	100.0%

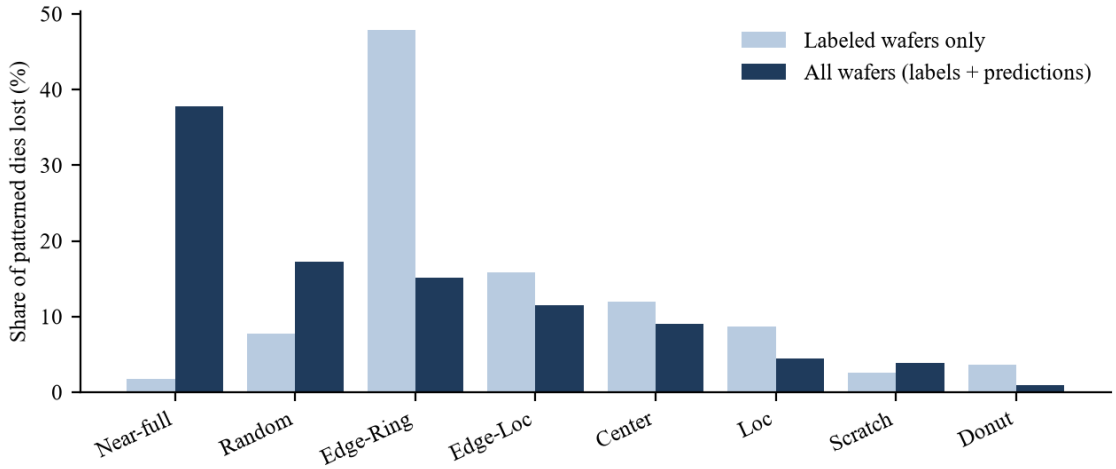


Figure 4. Share of patterned dies lost by pattern, among labeled wafers only and over all wafers.

5. Discussion

The two views of the data point to different priorities. On labeled wafers alone, Edge-Ring is the dominant patterned loss, which would argue for a sustained improvement effort on one chronic defect. Over all wafers, Near-full and Random lead. These are the same patterns most over-represented in flagged lots, which marks them as excursion patterns. They arrive in bursts, concentrate in specific lots, and cost hundreds of dies per affected wafer. The labeled-only analysis would have pointed effort at the wrong problem, and the check on label coverage is what revealed it.

For a program team, the results suggest three workstreams with different cadences and owners.

- **Excursion detection.** Review a Laney p′ chart of lot fail rate on a regular cadence, with flagged lots ranked by excess fails. The flagged lots here are 3.7% of lots but hold 24.2% of dies lost.
- **Containment for severe patterns.** Treat Near-full and Random wafers as triggers for prompt investigation of their lot, since each occurrence costs far more than a typical defect.
- **Chronic improvement.** Keep Edge-Ring as a longer-term improvement target. It remains a top-three loss in both views and is recognized reliably by the classifier, which makes progress

measurable.

This study identifies where loss concentrates, not why. Linking each pattern to a process step requires engineering data such as tool, chamber, and probe card history, which WM-811K does not include.

6. Limitations

- Lot order is inferred from lot numbers. The moving-range estimate in the Laney chart assumes this order reflects production sequence, which cannot be confirmed.
- Lots of different products are charted together. The positive relationship between lot size and yield (Spearman 0.34) suggests product mix contributes to the large σ_z. Charting by die count band as a stand-in for product would separate the two.
- The classifier is trained on labeled wafers, which differ from unlabeled ones. Held-out performance may overstate accuracy on unlabeled wafers, so the all-wafer Pareto is an estimate.
- Dies lost counts every failing die on a wafer, including background failures not caused by the labeled pattern, so pattern shares are upper bounds on the loss each pattern causes.

7. Conclusion

A reproducible pipeline over 811,457 wafer maps shows that a standard p-chart fails at this lot size, that a Laney p′ chart isolates 1,716 lots holding 24.2% of loss, and that the labeled-wafer Pareto misses the patterns driving most patterned loss. Extending labels with a classifier validated on held-out lots, and confirming the key result with a model-free check, moves the priority from chronic Edge-Ring loss to excursion patterns led by Near-full and Random. Code and outputs are available at github.com/michaelbaoo/wafer-yield.

References

[1] M.-J. Wu, J.-S. R. Jang, and J.-L. Chen, “Wafer map failure pattern recognition and similarity ranking for large-scale data sets,” *IEEE Transactions on Semiconductor Manufacturing*, vol. 28, no. 1, pp. 1–12, 2015.

[2] D. B. Laney, “Improved control charts for attributes,” *Quality Engineering*, vol. 14, no. 4, pp. 531–537, 2002.